10/8/2026

5. Experiment on Packet Capture tool Wireshark

Aim:- Experiments on Packet capture tool wireshark

Packet Sniffer:-

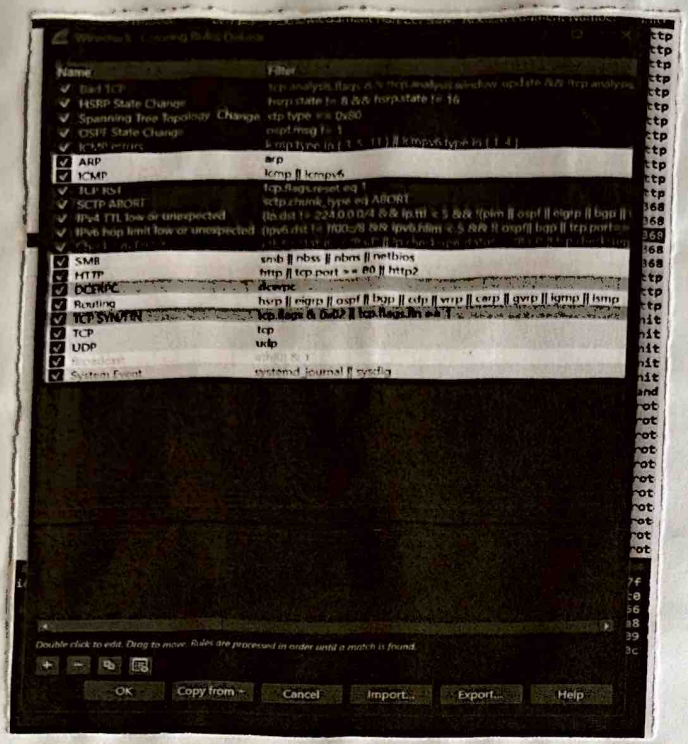
* Sniffs messages being sent/recieved from/by your computers
* Store and display the contents of the various protocol fields in the messages
* Passive Program
    i). Never sends packets itself
    ii). No packets addr to it
    iii). Receivers a copy of all packets (sent/received)

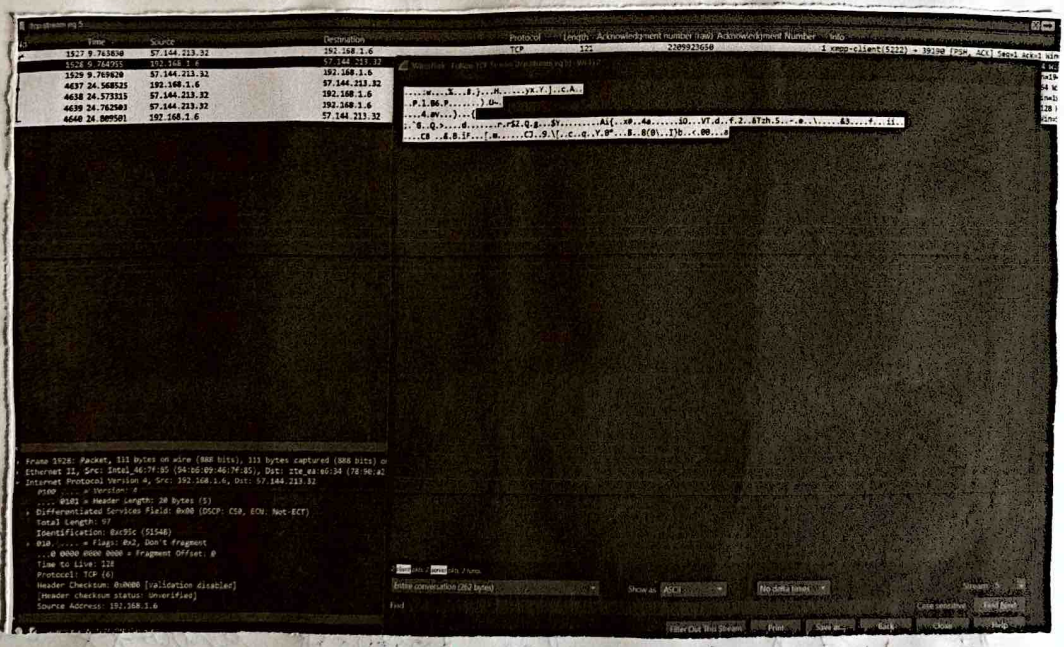
Packet Sniffer Structure Diagnostic tool:-

* Tcpdump - Eg:- tcpdump -enx host 10.129.41.2 -w exe3.out
* Wireshark - Wireshark -r exe3.out

Procedure:-

* Select the local area connection in Wireshark
* Go to capture option.


Name
Filter
ARP
arp
ICMP
icmp || icmpv6
SMB
smb || nbss || nbns || netbios
HTTP
http || tcp.port == 80 || http2
Routing
hsrp || eigrp || ospf || bgp || cdp || vrrp || carp || gvrp || igmp || ismp
TCP
tcp
UDP
udp
Double click to edit. Drag to move. Rules are processed in order until a match is found.
OK
Copy from ▾
Cancel
Import...
Export...
Help


Time
Source
Destination
Protocol
Length
Acknowledgment number (raw)
Acknowledgment Number
Info
1527 9.763830 57.144.213.32 192.168.1.6
1528 9.764795 192.168.1.6 57.144.213.32
1529 9.769820 57.144.213.32 192.168.1.6
4637 24.568525 192.168.1.6 57.144.213.32
4638 24.573315 57.144.213.32 192.168.1.6
4639 24.762503 57.144.213.32 192.168.1.6
4640 24.809501 192.168.1.6 57.144.213.32
TCP 121 2309923650 1 xmpp-client(5222) → 39190 [PSH, ACK] Seq=1 Ack=1 Win
Frame 1928: Packet, 111 bytes on wire (888 bits), 111 bytes captured (888 bits) o
Ethernet II, Src: Intel_46:7f:85 (94:e6:f7:46:7f:85), Dst: Ite_ea:66:34 (78:9d:a2
Internet Protocol Version 4, Src: 192.168.1.6, Dst: 57.144.213.32
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 97
Identification: 0xc95c (51548)
010. .... = Flags: 0x2, Don't Fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.1.6
Entire conversation (262 bytes)
Show as: ASCII
No delta times
Case sensitive
Find next
Find
Filter Out This Stream
Print
Save as...
Back
Close
Help

*. Select capture and stop automatically after 100 packets

*. Then click start capture

*. Save packets

i). Create a filter to display only TCP/UDP packets, inspect the packets and provide the flow graph.

*. Go to search bar at the top of the tool

*. Search TCP packets in search bar

*. To see flow graph click statistics flow graph

*. Save the packets

ii). Create a filter for ARP packets

*. Go to search option

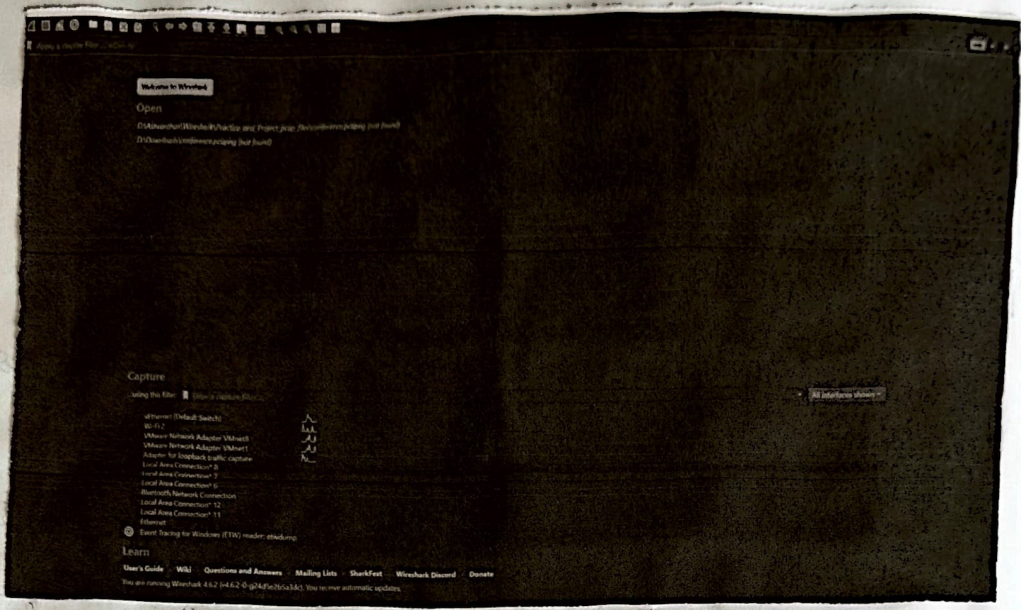
*. Search ARP packet in search bar

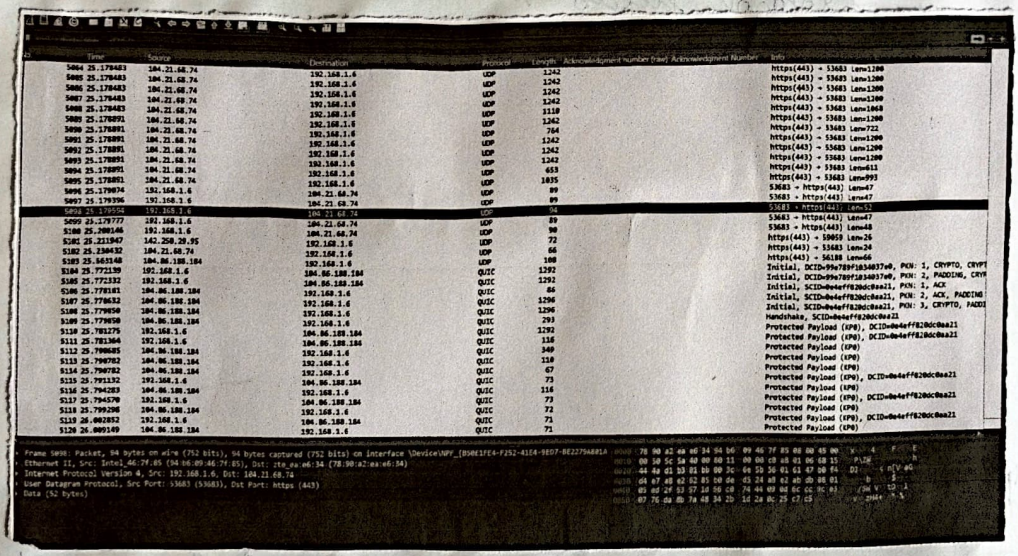
*. Save the packets

iii). Creating a filter to display DNS packets and provide the flow graph

iv). Creating a filter to display HTTP only packet and inspect the packet

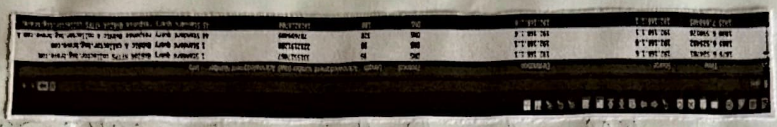
v). Create filter for ICMP and DHCP packets and inspect and show the packets.

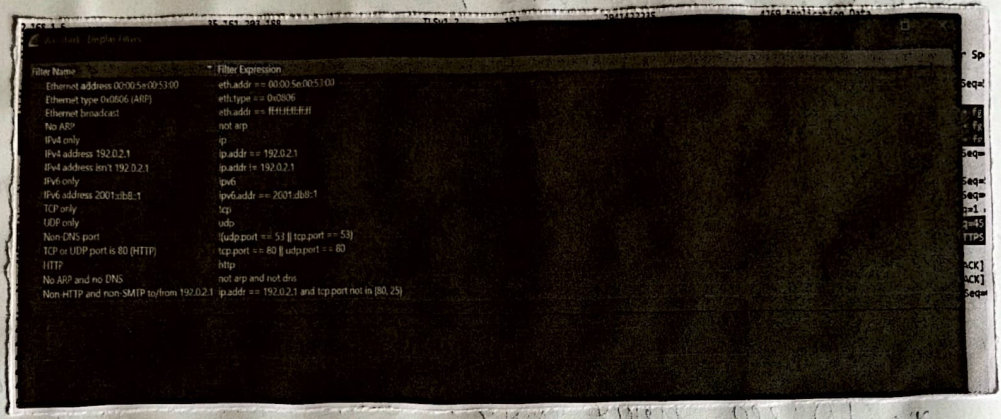

Welcome to Wireshark
Open
Capture
Learn
User's Guide
Wiki
Questions and Answers
Mailing Lists
SharkFest
Wireshark Discord
Donate


No. Time Source Destination Protocol Length Info
5064 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5065 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5066 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5067 25.178483 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5068 25.178483 104.21.68.74 192.168.1.6 UDP 1118 https(443) → 53683 Len=1068
5089 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5090 25.178891 104.21.68.74 192.168.1.6 UDP 764 https(443) → 53683 Len=722
5091 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5092 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5093 25.178891 104.21.68.74 192.168.1.6 UDP 1242 https(443) → 53683 Len=1200
5094 25.178891 104.21.68.74 192.168.1.6 UDP 653 https(443) → 53683 Len=611
5095 25.178891 104.21.68.74 192.168.1.6 UDP 1035 https(443) → 53683 Len=993
5096 25.179074 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5097 25.179396 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5098 25.179594 192.168.1.6 104.21.68.74 UDP 94 53683 → https(443) Len=52
5099 25.179777 192.168.1.6 104.21.68.74 UDP 89 53683 → https(443) Len=47
5100 25.200146 192.168.1.6 104.21.68.74 UDP 90 53683 → https(443) Len=48
5101 25.221947 142.250.20.95 192.168.1.6 UDP 72 https(443) → 59059 Len=26
5102 25.230432 104.21.68.74 192.168.1.6 UDP 66 https(443) → 53683 Len=24
5103 25.563148 104.86.188.184 192.168.1.6 UDP 108 https(443) → 56188 Len=66
5104 25.772139 192.168.1.6 104.86.188.184 QUIC 1292 Initial, DCID=99e789f1034037e0, PKN: 1, CRYPTO, CRYP
5105 25.772332 192.168.1.6 104.86.188.184 QUIC 1292 Initial, DCID=99e789f1034037e0, PKN: 2, PADDING, CRYP
5106 25.779101 104.86.188.184 192.168.1.6 QUIC 66 Initial, SCID=0e4eff820dc0ea21, PKN: 1, ACK
5107 25.779632 104.86.188.184 192.168.1.6 QUIC 1296 Initial, SCID=0e4eff820dc0ea21, PKN: 2, ACK, PADDING
5108 25.779950 104.86.188.184 192.168.1.6 QUIC 1296 Initial, SCID=0e4eff820dc0ea21, PKN: 3, CRYPTO, PADDI
5109 25.779950 104.86.188.184 192.168.1.6 QUIC 293 Handshake, SCID=0e4eff820dc0ea21
5110 25.781275 192.168.1.6 104.86.188.184 QUIC 1292 Protected Payload (KP0), DCID=0e4eff820dc0ea21
5111 25.781364 192.168.1.6 104.86.188.184 QUIC 116 Protected Payload (KP0), DCID=0e4eff820dc0ea21
5112 25.790685 104.86.188.184 192.168.1.6 QUIC 349 Protected Payload (KP0)
5113 25.790782 104.86.188.184 192.168.1.6 QUIC 110 Protected Payload (KP0)
5114 25.790782 104.86.188.184 192.168.1.6 QUIC 67 Protected Payload (KP0)
5115 25.791232 192.168.1.6 104.86.188.184 QUIC 73 Protected Payload (KP0), DCID=0e4eff820dc0ea21
5116 25.794283 104.86.188.184 192.168.1.6 QUIC 116 Protected Payload (KP0)
5117 25.794570 192.168.1.6 104.86.188.184 QUIC 77 Protected Payload (KP0), DCID=0e4eff820dc0ea21
5118 25.799298 104.86.188.184 192.168.1.6 QUIC 72 Protected Payload (KP0)
5119 26.002852 192.168.1.6 104.86.188.184 QUIC 71 Protected Payload (KP0), DCID=0e4eff820dc0ea21
5120 26.009149 104.86.188.184 192.168.1.6 QUIC 71 Protected Payload (KP0)
Frame 5098: Packet, 94 bytes on wire (752 bits), 94 bytes captured (752 bits) on interface \Device\NPF_{B5061EE4-F252-4164-9ED7-BE2279A881A
Ethernet II, Src: Intel_46:7f:85 (94:b6:09:46:7f:85), Dst: zte_ea:e6:34 (78:90:a2:ea:e6:34)
Internet Protocol Version 4, Src: 192.168.1.6, Dst: 104.21.68.74
User Datagram Protocol, Src Port: 53683 (53683), Dst Port: https (443)
Data (52 bytes)
0000 78 90 a2 ea e6 34 94 b6 09 46 7f 85 08 00 45 00
0010 00 50 5c 5a 40 00 80 11 00 00 c0 a8 01 06 68 15
0020 44 4a d1 b3 01 bb 00 3c 6e 5b 56 01 61 47 b0 f4
0030 d4 a7 48 e2 52 85 b0 de d5 24 a8 e2 ab db 88 01
0040 05 ad 2f 53 57 1d b6 c0 74 44 09 08 6c cc 9c 01
0050 07 76 da db 7a 48 34 2b 1d 2a 0c 25 c7 c5

Result :-

Wireshark helps us in analysing and understanding our network traffic and packet flow through various filters and flow graphs.




Capture Display Filters
Filter Name
Filter Expression
Ethernet address 00:00:5e:00:53:00
eth.addr == 00:00:5e:00:53:00
Ethernet type 0x0806 (ARP)
eth.type == 0x0806
Ethernet broadcast
eth.addr == ff:ff:ff:ff:ff:ff
No ARP
not arp
IPv4 only
ip
IPv4 address 192.0.2.1
ip.addr == 192.0.2.1
IPv4 address isn't 192.0.2.1
ip.addr != 192.0.2.1
IPv6 only
ipv6
IPv6 address 2001:db8::1
ipv6.addr == 2001:db8::1
TCP only
tcp
UDP only
udp
Non-DNS port
(udp.port == 53 || tcp.port == 53)
TCP or UDP port is 80 (HTTP)
tcp.port == 80 || udp.port == 80
HTTP
http
No ARP and no DNS
not arp and not dns
Non-HTTP and non-SMTP to/from 192.0.2.1
ip.addr == 192.0.2.1 and tcp.port not in {80, 25}